1.1.32 - Analog Signal Processing

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If $x = \text{Re } G(j\omega)$, and $y = \text{Im } G(j\omega)$ then for $\omega \to 0^+$, the Nyquist plot for

$$G(s) = \frac{1}{s(s+1)(s+2)}$$

becomes asymptotic to the line (EE 2007)

1. $x = 0$
2. $x = -3/4$
3. $x = y - 1/6$
4. $x = y/\sqrt{3}$

Theory

$G(s)$ represents the open-loop transfer function of a system in the Laplace domain.

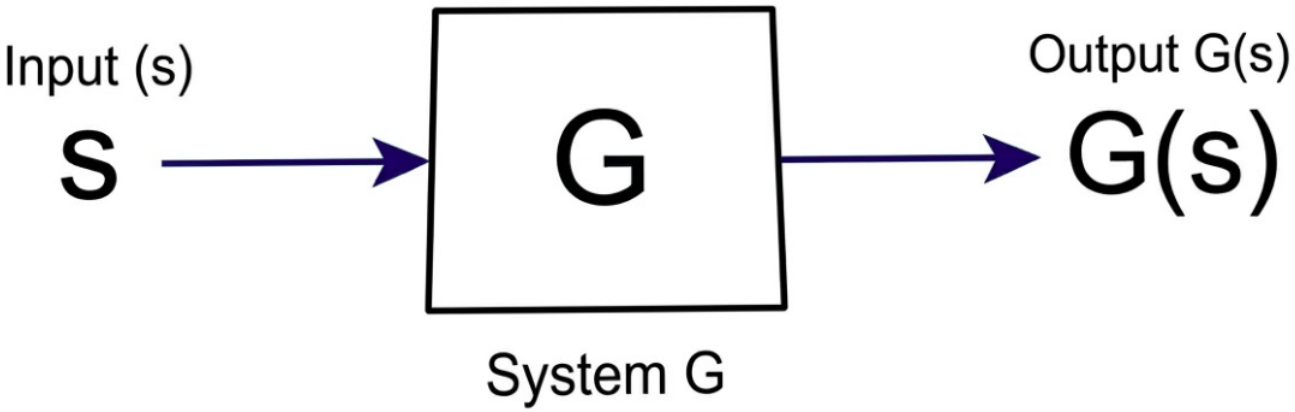


Figure: Function $G(s)$

$$G(s) = \frac{1}{s(s+1)(s+2)} \tag{1}$$

Transfer Function

When analyzing linear, time-invariant (LTI) systems in analog signal processing, working with differential equations in the time domain can get incredibly messy.By applying the Laplace transform (assuming all initial conditions are zero), we move from the time domain to the s-domain.

$$G(s) = \frac{\mathcal{L}\{\text{Output}\}}{\mathcal{L}\{\text{Input}\}} \tag{2}$$

Poles of System

From:

$$s(s+1)(s+2) = 0 \tag{3}$$

Poles are at :

- $s = 0$
- $s = -1$
- $s = -2$

Solution

Rewriting:

$$G(s) = \frac{1}{s} \cdot \frac{1}{(s+1)(s+2)} \tag{4}$$

- $\frac{1}{s} \rightarrow$ integrator
- $\frac{1}{(s+1)(s+2)} \rightarrow$ finite gain without discontinuity

Interpretation of Each Block

Integrator: $\frac{1}{s} = \frac{1}{j\omega}$

- Magnitude tends to infinity as $\omega \to 0$
- Phase is -90°
- Hence, the Nyquist plot is directed vertically downward near the origin

Remaining Block:

$$H(j\omega) = \frac{1}{(1 + j\omega)(2 + j\omega)} \tag{5}$$

At low frequencies ($\omega \to 0$):

- The system behaves as a constant gain element
- Evaluating at $\omega = 0$:

$$H(0) = \frac{1}{(1)(2)} = \frac{1}{2} \tag{6}$$

First-Order Approximation (Proof)

Given:

$$H(j\omega) = \frac{1}{(1+j\omega)(2+j\omega)} \tag{7}$$

Expand the denominator:

$$(1+j\omega)(2+j\omega) = 2 + 3j\omega + (j\omega)^2 \tag{8}$$

$$= 2 + 3j\omega - \omega^2 \tag{9}$$

For $\omega \to 0$, neglect higher-order terms:

$$(1+j\omega)(2+j\omega) \approx 2 + 3j\omega \tag{10}$$

Hence,

$$H(j\omega) \approx \frac{1}{2+3j\omega} \tag{11}$$

Factor out 2:

$$H(j\omega) = \frac{1}{2\left(1 + \frac{3}{2}j\omega\right)} \tag{12}$$

$$= \frac{1}{2} \cdot \frac{1}{1 + \frac{3}{2}j\omega} \tag{13}$$

Using the first-order approximation:

$$\frac{1}{1+x} \approx 1 - x \quad \text{for } |x| \ll 1 \tag{14}$$

Let $x = \frac{3}{2}j\omega$, then:

$$H(j\omega) \approx \frac{1}{2}\left(1 - \frac{3}{2}j\omega\right) \tag{15}$$

$$= \frac{1}{2} - j\frac{3}{4}\omega \tag{16}$$

$\therefore$

$$G(j\omega) = \frac{1}{j\omega} \cdot \left(\frac{1}{2} - j\frac{3}{4}\omega\right) \tag{17}$$

Hence,

$$x = \text{Re } G(j\omega) = \frac{-3}{4} \text{ and } y = \text{Im } G(j\omega) = \frac{-1}{2\omega} \tag{18}$$

Final Nyquist Behavior

$$x = \frac{-3}{4} \text{ and } y \to -\infty \tag{19}$$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

w = np.linspace(0.01, 1.0, 1000)

s = 1j * w
G = 1 / (s * (s + 1) * (s + 2))

x_exact = np.real(G)
y_exact = np.imag(G)

x_asymp = np.full_like(w, -0.75)
y_asymp = -1 / (2 * w)
```

Python Code

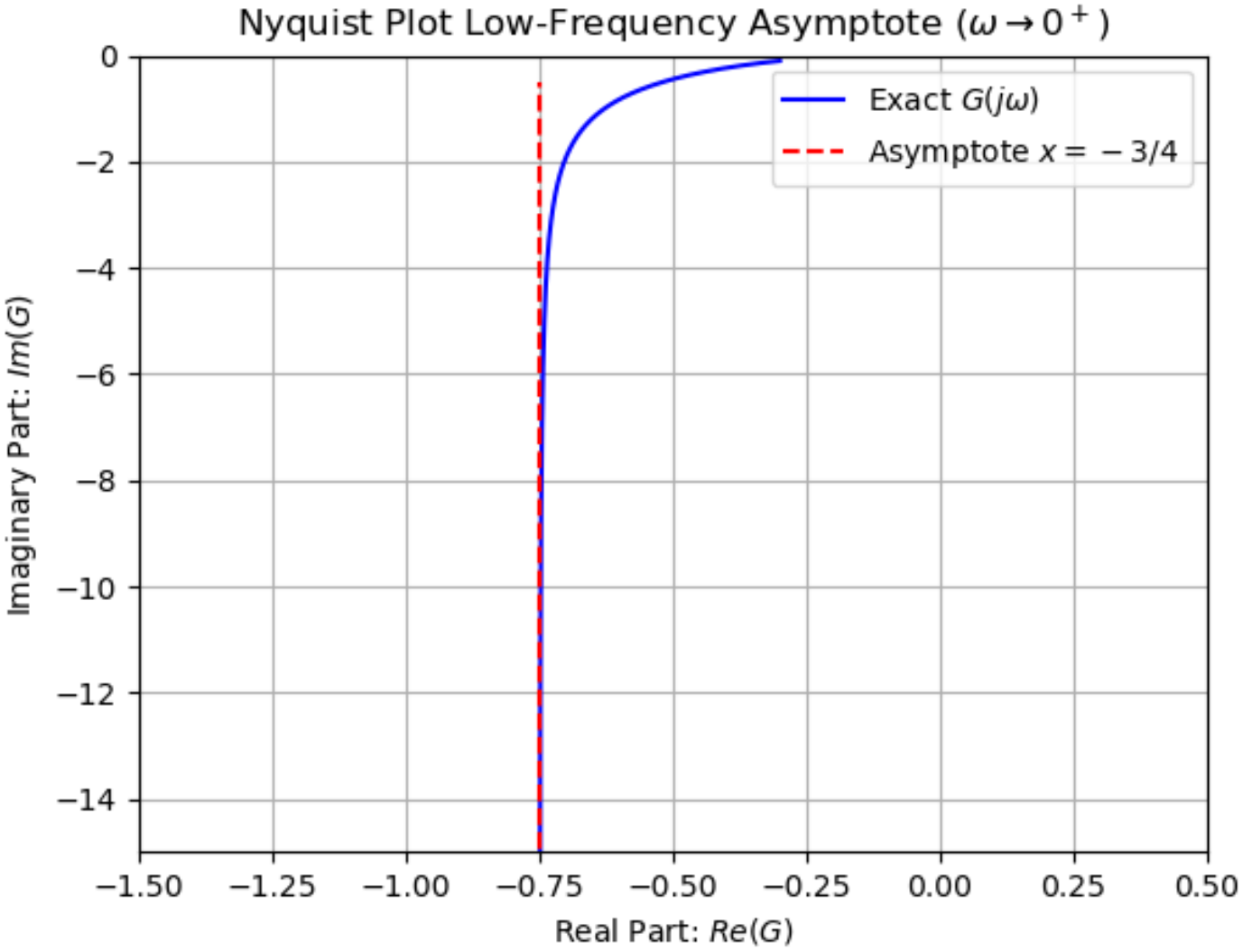
```
plt.figure()

plt.plot(x_exact, y_exact, label='Exact $G(j\omega)$', color='
    blue')
plt.plot(x_asymp, y_asymp, label='Asymptote $x = -3/4$', color='
    red', linestyle='--')

plt.title('Nyquist Plot Low-Frequency Asymptote ($\omega \\
    rightarrow 0^+$)')
plt.xlabel('Real Part: $Re(G)$')
plt.ylabel('Imaginary Part: $Im(G)$')
plt.grid()
plt.legend()
plt.xlim(-1.5, 0.5)
plt.ylim(-15, 0)

plt.show()
plt.savefig("../fig/plt1.png")
```

Plot


Nyquist Plot Low-Frequency Asymptote ($\omega \to 0^+$)
Exact $G(j\omega)$
Asymptote $x = -3/4$
Imaginary Part: $Im(G)$
Real Part: $Re(G)$
0
−2
−4
−6
−8
−10
−12
−14
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50